Stern-Gerlach with Qiskit

Day 1 · Sunday, Oct 11
MN Saha Auditorium
IISER Kolkata



A beam in,
two spots
out

The 1922 experiment

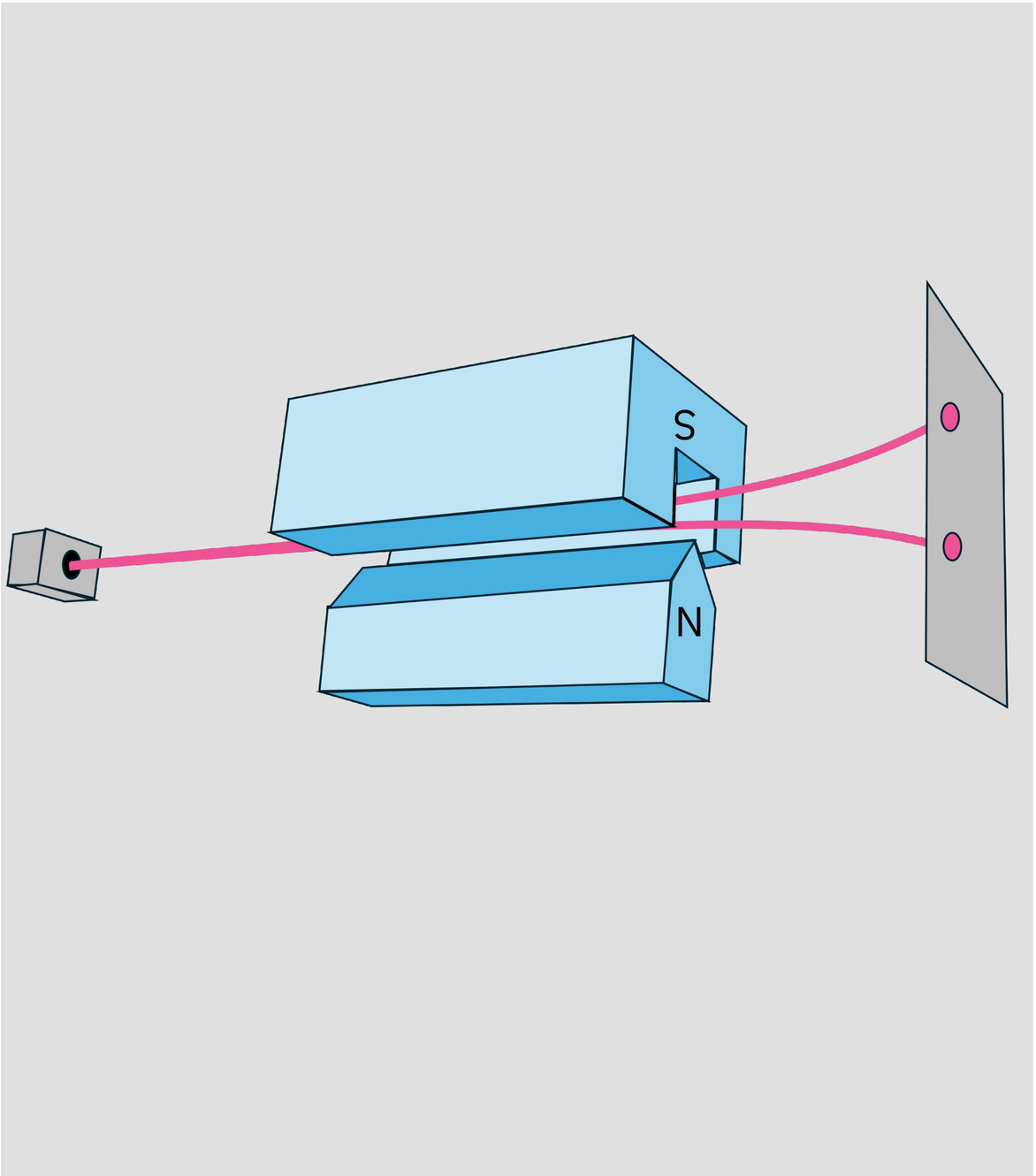
- Silver atoms are heated in an oven.

- A collimated beam passes through an inhomogeneous magnetic field.

- A screen records where the atoms land.

The goal: test whether angular momentum is quantized.

Orbital angular momentum was the target. Spin was proposed only in 1925.



Why a field gradient deflects atoms

Angular momentum gives a magnetic moment.

A field gradient along z exerts a force on that moment.

So the deflection measures the z-component of angular momentum.

If the z-component is quantized, the deflection is quantized too.

$$\vec{\mu} = \frac{gq}{2m}\vec{J}$$

$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

$$\vec{F} = \nabla(\vec{\mu} \cdot \vec{B})$$

$$F_z = \mu_z \frac{dB_z}{dz}$$

$$F_z = \frac{gq}{2m}\frac{dB_z}{dz}J_z$$



Classical Picture

Line

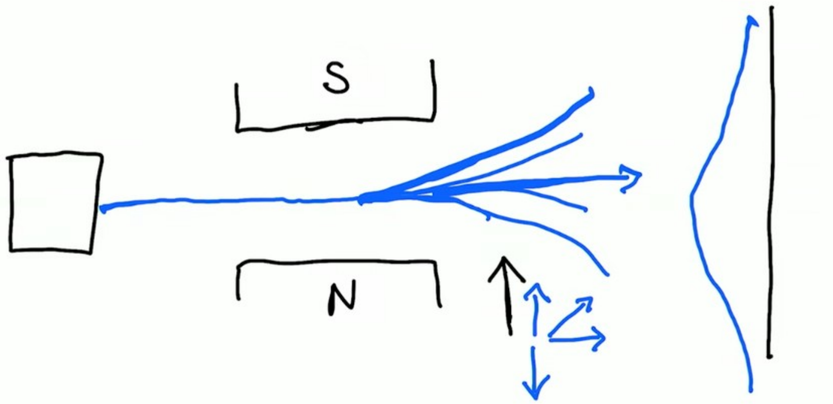
Quantum Picture

2

Source: Stern and Gerlach, 1922

Classical Picture

Classical magnets with random orientations would smear into a continuous line.



Line

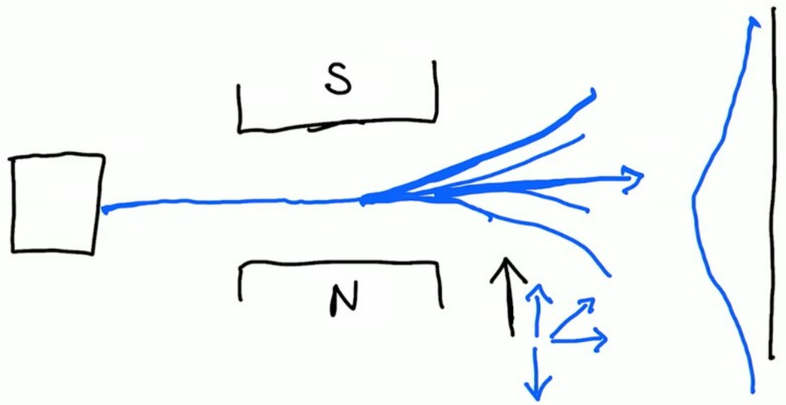
Quantum Picture

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Source: Stern and Gerlach, 1922

Classical Picture

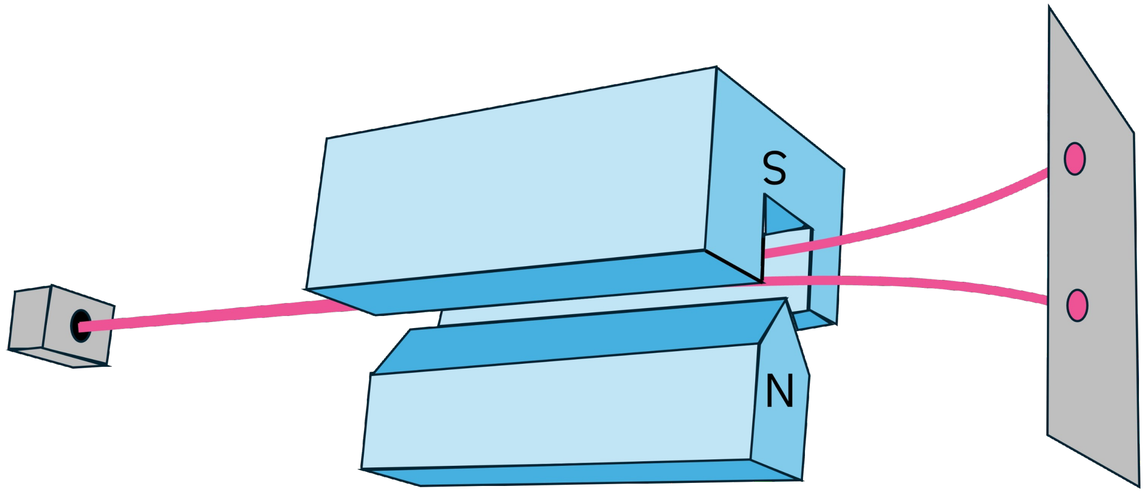
Classical magnets with random orientations would smear into a continuous line.



Line

Quantum Picture

Stern and Gerlach saw atoms land in only two places.



2

Source: Stern and Gerlach, 1922

From atoms to qubits

Spin-1/2 atom	Qubit
Spin up or spin down along the field axis. One measurement gives one of the two outcomes.	State 0 or 1 along z. One measurement gives one of the two outcomes. We map up to 0 and down to 1.



Four experiments on z, then one on x

1

Single measurement. One shot gives 0 or 1, never in between.

2

Many shots. The state sets the probabilities.

3

Random spins. An oven of random orientations.

4

Measure twice. The second result repeats the first.

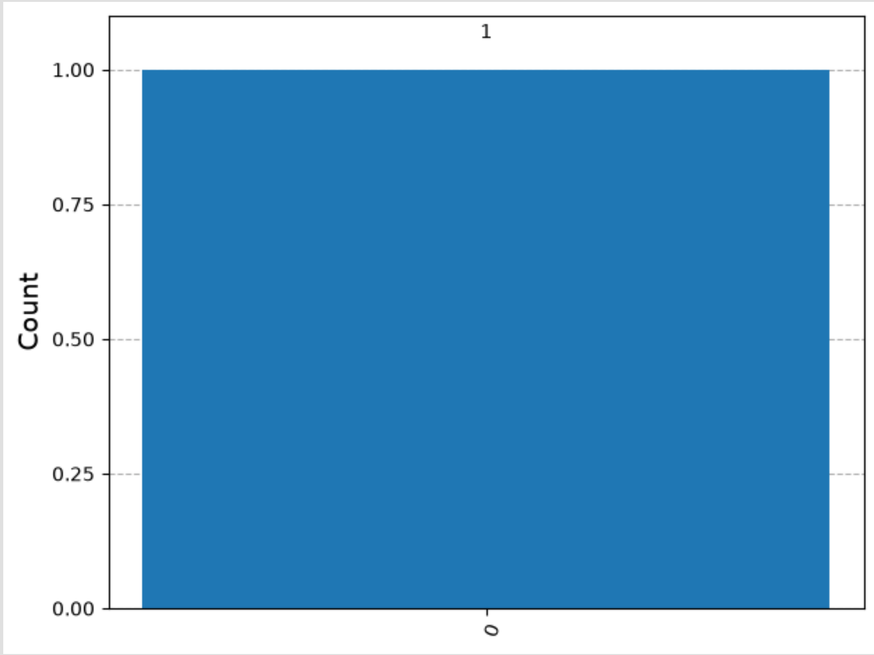
Experiments 1 and 2: one shot, many shots

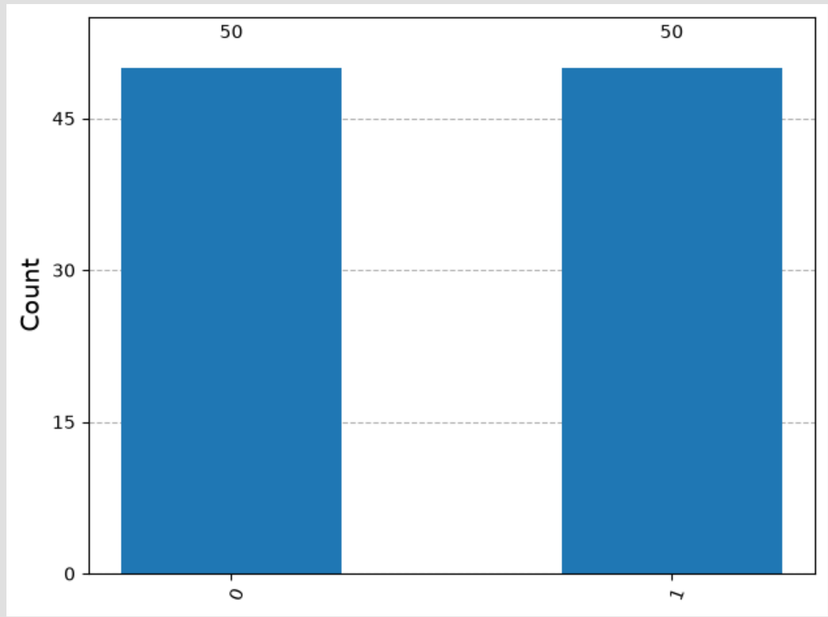
Prepare a random state $a|0\rangle + b|1\rangle$.

Measure once: the result is 0 or 1.

Measure 100 times: both outcomes appear.

The state sets the probabilities. Each single shot is random.



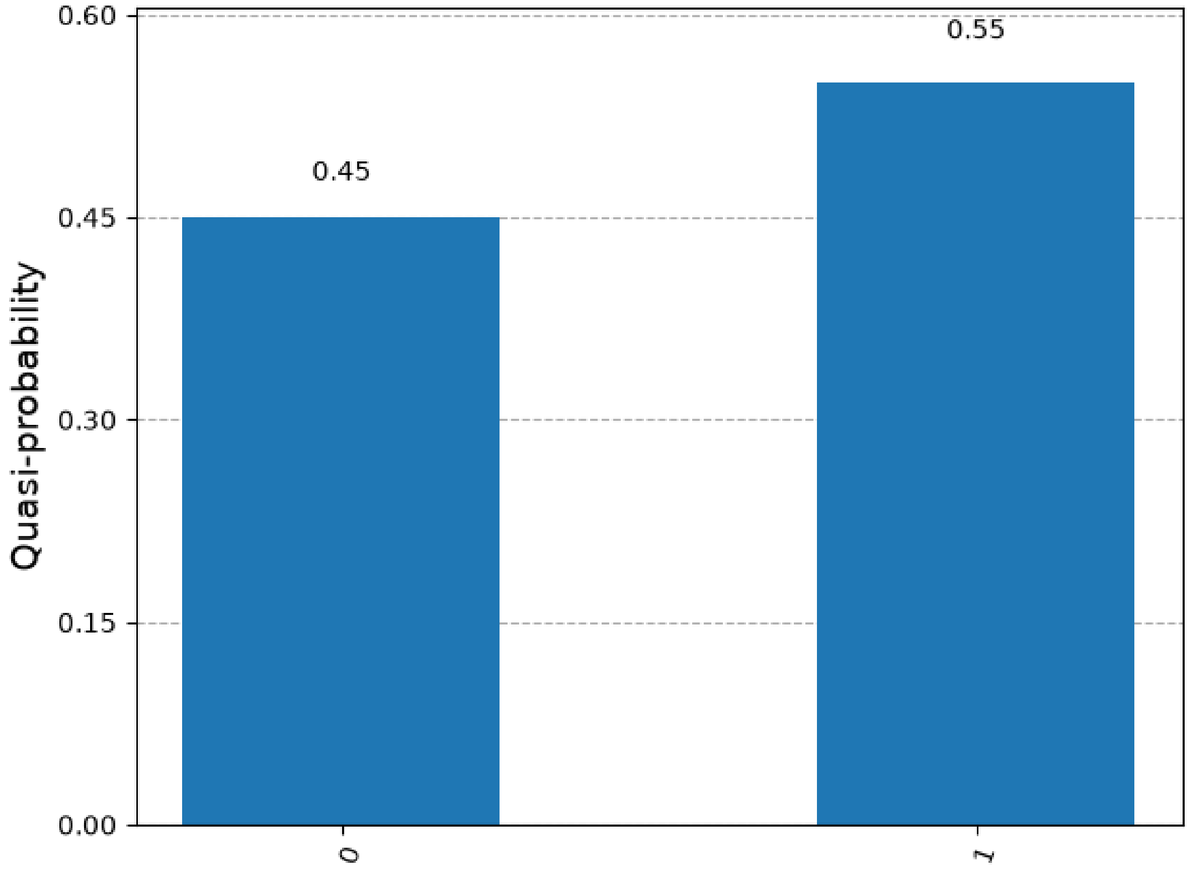


Experiment 3: random spins in an oven

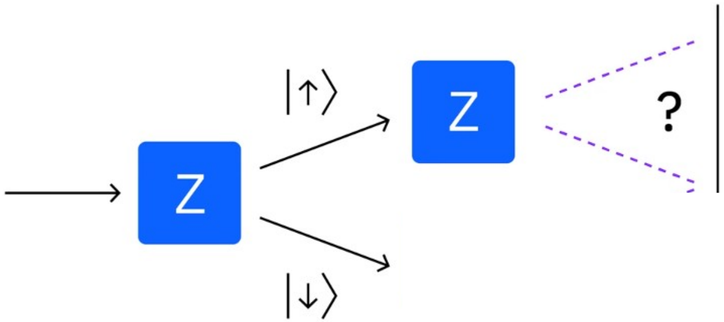
Random angles theta and phi give each atom a random orientation.

Measure each atom once.

We get about half 0 and half 1, like the original experiment.



Measure the same qubit twice along z: the second result matches the first.



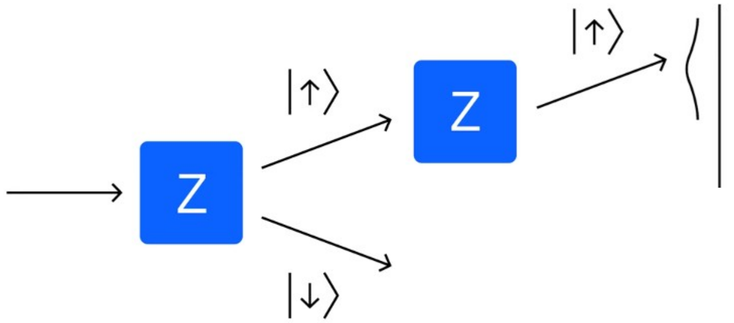
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Outcomes never seen on this ideal simulator.

01, 10

Real hardware shows about 1% flips from noise

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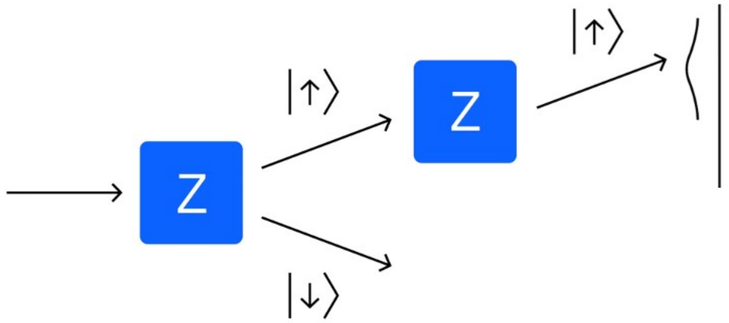
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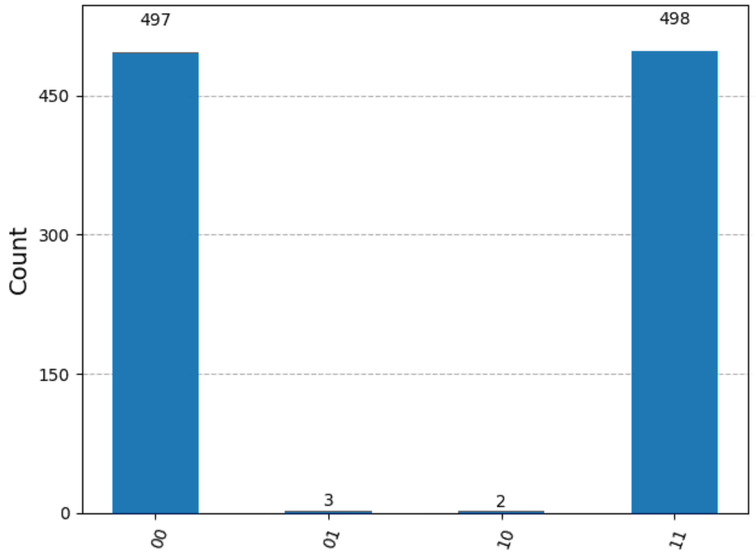
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Noise on Real Hardware



01, 10

Real hardware shows about 1% flips from noise

From observations to state vectors

Repeated measurements fix the vectors for up and down.

The measured S_z values are +ħ/2 and −ħ/2.

Operator action is not measurement.

Measuring always gives one eigenvalue and collapses the state.

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$S_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

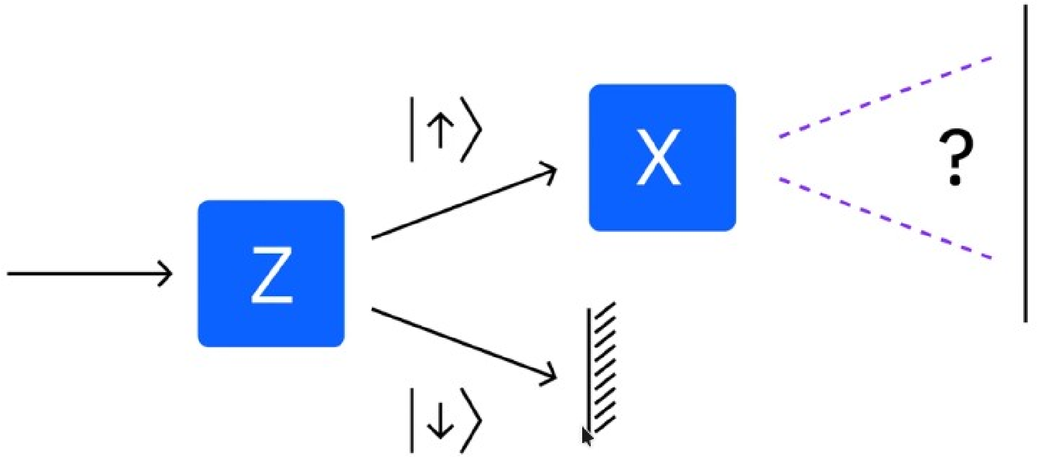
Experiment 5: measuring along x

Qubits only measure along z.

A Hadamard gate changes basis, so a z-measurement reveals x.

Start in $|0\rangle$: 50-50 for plus or minus x.

Measure x twice: the results repeat.



What we learned

1

Two outcomes. Measuring a two-level system gives one of two allowed values.

2

Random states. Either outcome is possible, with probabilities set by the state.

3

Repeat and rotate. Repeating a measurement repeats the result. Measuring along another axis is 50-50.

Adapted from IBM Qiskit in Classrooms: Stern-Gerlach measurements with Qiskit

Now, open the Stern-Gerlach notebook

